

MARK KESSLER

Senior Software Engineer · High-Performance Python & Scientific Computing

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LinkedIn | Portfolio | GitHub | PyPI: phynetpy

SUMMARY

Software engineer with 6 years building high-performance Python systems, the last four as technical lead. I own architecture, performance, CI/CD, releases, and documentation for PhyNetPy, an open-source library on PyPI, and I set technical direction, run breaking API migrations, and write the conventions other contributors build against. Most of my work is on graph and statistical inference problems where the straightforward implementation is far too slow, so I spend a lot of time profiling and then fixing what the profiler actually points at. I'm equally comfortable shipping the full stack around those systems: Flask/FastAPI services, REST APIs, and framework-free frontends for interactive data views.

TECHNICAL SKILLS

Languages

Python, Cython, JavaScript, TypeScript, C, Java, C#/ .NET, SQL

Performance

Profiling (cProfile, tracemalloc), Cython extension modules, caching, vectorization, multiprocessing, CUDA/CuPy GPU offload

Data & ML

NumPy, SciPy, PyTorch, TensorFlow, scikit-learn, pandas, NetworkX

Web & Backend

ES modules, Canvas/SVG rendering, React, Flask, FastAPI, REST API design, SQLite, PostgreSQL, MongoDB

Infrastructure

GitHub Actions CI/CD, PyPI packaging and release automation, Docker, pytest, Git

AI-Assisted Dev

Cursor, Claude Code, LangChain, Agentic Development with sandboxing and human approval

EXPERIENCE

Senior Software Engineer — Technical Lead, PhyNetPy

August 2020 – Present

Rice University, Computational Systems Lab — Houston, TX (Remote)

TECHNICAL LEADERSHIP & OWNERSHIP

- Technical lead and principal architect of PhyNetPy, an open-source phylogenetic network library on PyPI. Wrote 217 of its 249 commits, covering architecture, implementation, packaging, docs, and user support.
- Led the 0.6.0 API redesign, replacing an overgrown command surface with a typed three-axis API (data × model × criterion) behind three verbs. Wrote the migration guide and compatibility matrix so users could move across a breaking release without redoing work.
- Wrote the coding standards and repo-level rules that steer AI coding assistants on import structure, naming, type hints, docstrings, and validation patterns, so generated code actually fits the codebase. Generated changes get reviewed against the same bar as human ones.
- Own CI/CD: a GitHub Actions matrix across Python 3.9–3.14, cross-platform smoke tests, and wheel install validation gating every release.
- Grew the test suite to **1,130 tests** at a **100% pass rate**, including memory regression, stress, and numerical cross-checks against the incumbent Java implementation.

- Built the release and docs tooling: a one-command deploy pipeline (version bump → test → build → PyPI) and a 1,100-line AST-based API reference generator.
- Onboarded and reviewed collaborators' contributions, wrote the NSF Year-1 technical report, and worked with the principal investigator on the technical roadmap.

PERFORMANCE ENGINEERING

- Shipped four Cython extension modules (graph core, pseudo-likelihood DP, coalescent DP, sequence kernels). The pseudo-likelihood kernel runs **up to 9.2x** faster than the pure-Python version and returns identical results.
- Replaced generic matrix exponentiation with closed-form and cached-eigendecomposition transition matrices across eight substitution models: **8.6–19.5x** faster than `scipy.linalg.expm`, matching it to $\sim 1e-16$.
- Rebuilt network adjacency on Cython O(1) index maps, giving **~11x** faster in-edge lookup and **~3.8x** faster leaf enumeration than NetworkX on the same topologies.
- Raised search throughput **2.8x** (2,742 → 972 ms/iteration) after profiling turned up 3.7M redundant graph traversals, which I replaced with memoized lookups.
- Cut topological error **2.9x** against the incumbent PhyloNet implementation at comparable runtime, and ran **7.4x** faster (110 s vs. 818 s) on the hardest benchmark case.
- Added GPU likelihood scoring with CuPy sparse contractions and VRAM-aware batching, so out-of-memory conditions surface as recoverable errors instead of killing the run. Also added multiprocessing scoring pools and parallel MCMC chains.
- Built Python machine-learning pipelines for large-scale data processing and model experimentation.
- Implemented tensor operations and data-encoding pipelines for high-volume numerical workloads.
- Experimented with different Neural Network architectures and probed for optimal model hyperparameters for inferring quartet networks from large DNA data sets.

PROJECTS

PhyNetPy — *v0.6.0 · Python 3.9–3.14 · ~41k lines across 49 modules + 4 Cython extensions*

- Framework for phylogenetic network inference, simulation, and comparison, with a typed public API, registry-based method dispatch, and explicit branch-length units that rule out a class of silent scientific errors. Sole owner across six public releases.

BioStore — *JavaScript · Flask · SQLite · Docker*

- Full-stack platform for submitting, monitoring, and visualizing long-running PhyNetPy jobs.
- Wrote the frontend by hand: ~2,500 lines of framework-free ES modules covering interactive network rendering, alignment views, a file browser, an in-browser editor, and the API client layer.
- Backend handles token auth, job execution, and lease-based file locking on SQLite (WAL) so multiple users can edit the same dataset without clobbering each other.
- This is an experiment in AI/Agentic development best practices. I currently use Cursor/Claude, bounded by codebase wide rulesets and OpenHands to regulate the workflow.

NBC-MAK: DFW and Tropical Weather Forecasting — *Local AI model training and CUDA development*

- My passion project! I enjoy tracking tropical systems, and these complex weather time series datasets are perfect for learning how to locally train AI models and work with large data.
- Always-on public API data pulling, and always-on local training.
- Responsive React front end, with beautiful map overlays, data visualization, data sandboxing, and statistical analyses of my local model compared with 6+ other large scale public models.
- CUDA enabled development, dealing with vast tensors of atmospheric data across time.

EDUCATION

Rice University — Bachelor of Arts, Computer Science

GPA 3.4